



efrei

PARIS PANTHÉON-ASSAS UNIVERSITÉ

Projet ArtConnect

Livrable 2

Groupe 11

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1. Dépendances fonctionnelles

Artist

id_artist -> name, bio, birthYear, contactEmail, phone, city, website, socialMedia, isActive

Artwork

id_artwork -> title, creationYear, type, medium, dimensions, description, price, status,
id_artist

Gallery

id_gallery -> name, address, ownerName, openingHours, contactPhone, rating, website

Exhibition

id_exhibition -> title, startDate, endDate, description, curatorName, theme, id_gallery

Workshop

id_workshop -> title, date, durationMinutes, maxParticipants, price, location, description,
level, id_artist

CommunityMember

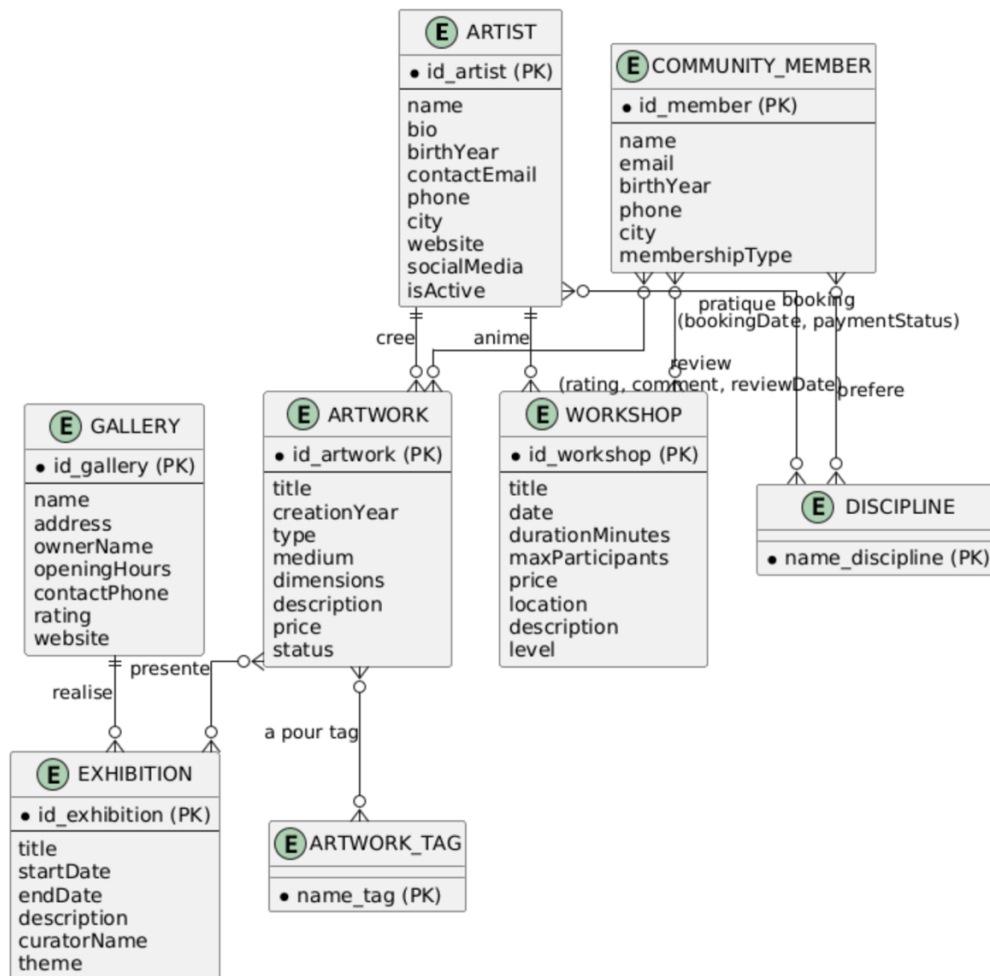
id_member -> name, email, birthYear, phone, city, membershipType

Associations

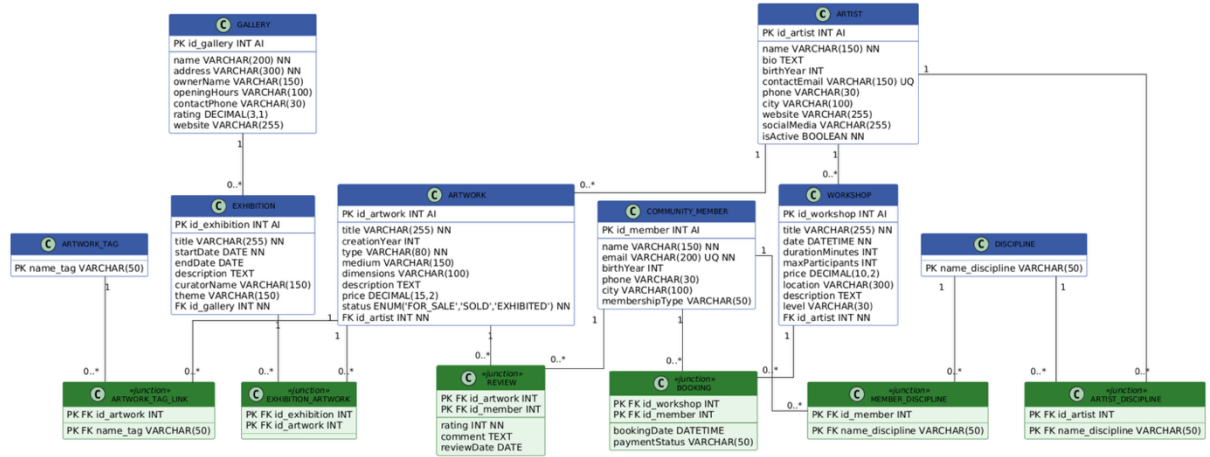
Review: (id_artwork, id_member) -> rating, comment, reviewDate

Booking: (id_workshop, id_member) -> bookingDate, paymentStatus

2. modèle Conceptuel de Données (ERD)



3. modèle Logique de Données (MLD)



4. Normalisation

Première Forme Normale (1FN)

Condition : attributs atomiques, pas de groupes répétitifs, unicité des lignes.

- Tous les attributs sont atomiques : name, email, city, etc. sont des valeurs simples.
- Les attributs multivalués (disciplines, tags) sont extraits dans des tables séparées avec tables de jonction.
- Chaque table possède une clé primaire INT AUTO_INCREMENT garantissant l'unicité.

Résultat : toutes les tables respectent la 1FN.

Deuxième Forme Normale (2FN)

Condition : être en 1FN et que chaque attribut non-clé dépend de la totalité de la clé.

- Tables à clé simple (id) : 2FN automatiquement satisfaite.
- Tables de jonction BOOKING et REVIEW : les attributs (bookingDate, paymentStatus, rating, comment, reviewDate) dépendent bien de la paire complète (id_workshop + id_member) ou (id_artwork + id_member), pas d'une seule partie.
- Tables de jonction pures (ARTIST_DISCIPLINE, EXHIBITION_ARTWORK...) : pas d'attributs non-clés donc 2FN triviale.

Résultat : toutes les tables respectent la 2FN.

Troisième Forme Normale (3FN)

Condition : être en 2FN et aucun attribut non-clé ne dépend transitivement d'un autre attribut non-clé.

- ARTWORK contient id_artist (FK) mais pas le nom de l'artiste -> aucune dépendance transitive.
- EXHIBITION contient id_gallery (FK) mais pas le nom de la galerie -> aucune dépendance transitive.
- WORKSHOP contient id_artist (FK) mais pas le nom de l'instructeur -> aucune dépendance transitive.
- Aucun attribut non-clé ne détermine un autre attribut non-clé dans la même table.

Résultat : toutes les tables respectent la 3FN.

5. Script SQL de création de la base

```
1 CREATE DATABASE IF NOT EXISTS artconnect;
2 USE artconnect;
3
4 CREATE TABLE ARTIST (
5     id_artist INT AUTO_INCREMENT PRIMARY KEY,
6     name VARCHAR(150) NOT NULL,
7     bio TEXT,
8     birthYear INT,
9     contactEmail VARCHAR(150) UNIQUE,
10    phone VARCHAR(30),
11    city VARCHAR(100),
12    website VARCHAR(255),
13    socialMedia VARCHAR(255),
14    isActive BOOLEAN NOT NULL DEFAULT TRUE
15 );
16
17 CREATE TABLE DISCIPLINE (
18     name_discipline VARCHAR(50) PRIMARY KEY
19 );
20
21 CREATE TABLE ARTIST_DISCIPLINE (
22     id_artist INT NOT NULL,
23     name_discipline VARCHAR(50) NOT NULL,
24     PRIMARY KEY (id_artist, name_discipline),
25     FOREIGN KEY (id_artist) REFERENCES ARTIST(id_artist) ON DELETE CASCADE,
26     FOREIGN KEY (name_discipline) REFERENCES DISCIPLINE(name_discipline) ON DELETE CASCADE
27 );
28
29 CREATE TABLE ARTWORK_TAG (
30     name_tag VARCHAR(50) PRIMARY KEY
31 );
32
33 CREATE TABLE ARTWORK_TAG_LINK (
34     id_artwork INT NOT NULL,
35     name_tag VARCHAR(50) NOT NULL,
36     PRIMARY KEY (id_artwork, name_tag),
37     FOREIGN KEY (id_artwork) REFERENCES ARTWORK(id_artwork) ON DELETE CASCADE,
38     FOREIGN KEY (name_tag) REFERENCES ARTWORK_TAG(name_tag) ON DELETE CASCADE
39 );
40
41 CREATE TABLE GALLERY (
42     id_gallery INT AUTO_INCREMENT PRIMARY KEY,
43     name VARCHAR(200) NOT NULL,
44     address VARCHAR(300) NOT NULL,
45     ownerName VARCHAR(150),
46     openingHours VARCHAR(100),
47     contactPhone VARCHAR(30),
48     rating DECIMAL(3,1),
49     website VARCHAR(255)
50 );
51
52 CREATE TABLE EXHIBITION (
53     id_exhibition INT AUTO_INCREMENT PRIMARY KEY,
54     title VARCHAR(255) NOT NULL,
55     startDate DATE NOT NULL,
56     endDate DATE,
57     description TEXT,
58     curatorName VARCHAR(150),
59     theme VARCHAR(150),
60     id_gallery INT NOT NULL,
61     FOREIGN KEY (id_gallery) REFERENCES GALLERY(id_gallery) ON DELETE RESTRICT
62 );
```

```

79 • CREATE TABLE EXHIBITION_ARTWORK (
80     id_exhibition INT NOT NULL,
81     id_artwork     INT NOT NULL,
82     PRIMARY KEY (id_exhibition, id_artwork),
83     FOREIGN KEY (id_exhibition) REFERENCES EXHIBITION(id_exhibition) ON DELETE CASCADE,
84     FOREIGN KEY (id_artwork) REFERENCES ARTWORK(id_artwork) ON DELETE CASCADE
85 );
86
87 • CREATE TABLE WORKSHOP (
88     id_workshop      INT AUTO_INCREMENT PRIMARY KEY,
89     title             VARCHAR(255) NOT NULL,
90     date              DATETIME NOT NULL,
91     durationMinutes  INT,
92     maxParticipants  INT,
93     price             DECIMAL(10,2),
94     location          VARCHAR(300),
95     description       TEXT,
96     level             VARCHAR(30),
97     id_artist         INT NOT NULL,
98     FOREIGN KEY (id_artist) REFERENCES ARTIST(id_artist) ON DELETE RESTRICT
99 );
100
101 • CREATE TABLE COMMUNITY_MEMBER (
102     id_member        INT AUTO_INCREMENT PRIMARY KEY,
103     name             VARCHAR(150) NOT NULL,
104     email            VARCHAR(200) NOT NULL UNIQUE,
105     birthYear        INT,
106     phone            VARCHAR(30),
107     city             VARCHAR(100),
108     membershipType   VARCHAR(50)
109 );
110
111
112 • CREATE TABLE MEMBER_DISCIPLINE (
113     id_member         INT NOT NULL,
114     name_discipline   VARCHAR(50) NOT NULL,
115     PRIMARY KEY (id_member, name_discipline),
116     FOREIGN KEY (id_member) REFERENCES COMMUNITY_MEMBER(id_member) ON DELETE CASCADE,
117     FOREIGN KEY (name_discipline) REFERENCES DISCIPLINE(name_discipline) ON DELETE CASCADE
118 );
119
120 • CREATE TABLE BOOKING (
121     id_workshop INT NOT NULL,
122     id_member   INT NOT NULL,
123     bookingDate DATETIME,
124     paymentStatus VARCHAR(50),
125     PRIMARY KEY (id_workshop, id_member),
126     FOREIGN KEY (id_workshop) REFERENCES WORKSHOP(id_workshop) ON DELETE CASCADE,
127     FOREIGN KEY (id_member)   REFERENCES COMMUNITY_MEMBER(id_member) ON DELETE CASCADE
128 );
129
130 • CREATE TABLE REVIEW (
131     id_artwork INT NOT NULL,
132     id_member  INT NOT NULL,
133     rating     INT NOT NULL CHECK (rating BETWEEN 1 AND 5),
134     comment    TEXT,
135     reviewDate DATE,
136     PRIMARY KEY (id_artwork, id_member),
137     FOREIGN KEY (id_artwork) REFERENCES ARTWORK(id_artwork) ON DELETE CASCADE,
138     FOREIGN KEY (id_member)   REFERENCES COMMUNITY_MEMBER(id_member) ON DELETE CASCADE
139 );

```